Tab Title: Snake Game Elements

What does the player do? What is the goal?

The player controls a growing line (the "snake"), which moves around on a two-dimensional grid. The goal is to steer the snake so that it eats randomly placed food items. Each food item gives a point and makes the snake grow longer by one segment.

How does the player win?

The game is traditionally an endless score competition, with no real win condition apart from getting the highest score. Another way to win is to fill the game board completely with the snake.

How does the player lose?

The game ends abruptly when the player loses. The player loses by running the snake's head into any of the four outer walls of the game board or into the snake's own body.

What keys, buttons, or mouse actions work?

Up Arrow / W: Snake moves up
Down Arrow / S: Snake moves down
Left Arrow / A: Snake moves left
Right Arrow / D: Snake moves right
The snake cannot reverse direction completely (e.g., move down when going up). This prevents the player from hitting himself immediately after changing direction.
Where can the player move? What are the boundaries?
Movement occurs on a 2D grid of a fixed size within the game window. The entire snake must be kept inside the grid. The moment any part of the snake's head hits the outer wall of the grid, the game ends in a loss.

What is on the screen? Any UI or user interface elements?

Header / Score Board: Shows current score and high score
Game Arena: The playing field, with the snake made out of tiles and food appearing as a special tile.
Overlay Screen: Shows a "Game Over" message, along with the score, the high score, and a way to restart the game (e.g., a button or a message telling the player to press Space to continue).

How does the player's score change? How does the player earn or lose points?

The player gains +10 points every time the snake eats food. The score resets upon losing. The score is kept in a variable, which is increased by 10 every game loop iteration, when the snake's head overlaps with the food's position.

Title: My Game

Game Name: Retro Pong

Function: What does the player do? What is the goal?

Two players should use their vertical paddles placed on either side of the screen to bounce the ball back at their opponent. The player must maneuver his or her paddle to allow the ball to pass through his or her opponent's side of the screen to make him or her score a point. The game takes place on a rectangular screen where two paddles and a small ball are visible. The screen has a black background with a gray dashed line dividing it into two halves.

Win Condition: How does the player win?

The first player (Left or Right) to reach ten points wins the game. After winning, the game will announce the victory with a victory screen for five seconds before restarting.

Lose Condition: How does the player lose?

The player loses a round when the ball passes their paddle on their side and scores them one point. The player can also lose a match when their opponent reaches 10 points before them.

Controls: What keys, buttons, or mouse action?

Left Player:W – Up, S – Down

Right Player: UP Arrow – Up, DOWN Arrow – Down

Boundaries: Where can the player move? What limits apply?

Paddles can only move vertically on the left and right sides of the screen. The Paddles have a limitation of the top ($y = 0$) and bottom ($y = 500$) edges of the window.

Balls bounce off the top and bottom walls. The ball passes the left ($x < 0$) and right ($x > 700$) walls when it goes beyond their edges. This action adds a point to the opponent and resets the ball at the center of the screen.

UI or user interface: What appears on the screen?

Scoreboard: Display of points of the Left Player and Right Player at the top of the screen.

The center divider refers to a gray dashed line that runs from top to bottom in the middle of the screen.
The game elements consist of a black background with two white rectangular paddles and a circular ball.
The victory screen displays a text “Left Player Won!” or “Right Player Won!” depending on which player won the match in the middle of the screen.

Scoring: How does the player earn or lose points?

The game adds a point to the Right Player when the ball passes to the left of the left paddle ($x < 0$). It also adds a point to the Left Player when the ball passes to the right of the right paddle ($x > 700$). The ball’s angle changes depending on where it collided with the paddle, in relation to the center of the paddle.

Title: My AI Prompt

Create a complete, single-file 2D table tennis game called "Retro Pong".

Game Elements & Specifications:
1. Function: A 2-player paddle game where players bounce a ball back and forth across the screen to score points past the opponent's paddle.
2. Win Condition: The first player to reach 10 points wins the game, displaying a victory message ("Left Player Won!" or "Right Player Won!") before resetting.
3. Lose Condition: A player loses a point when the ball passes their paddle and exits their side of the screen.
4. Controls: 2-player local keyboard controls. Left player uses 'W' (Up) and 'S' (Down). Right player uses 'UP Arrow' (Up) and 'DOWN Arrow' (Down).
5. Boundaries: Paddles are restricted to vertical movement within the screen height (700x500 area). The ball bounces off top and bottom boundaries and resets to the center after a point is scored.

6. UI: Score display for both players at the top, a dashed center dividing line, white paddles and ball on a black background, and a centered winner text overlay upon match completion.
7. Scoring: +1 point awarded to the opponent whenever the ball passes a player's paddle. Ball deflection angle should vary based on impact point on the paddle.

Technical Requirements:
Use HTML, CSS, and JavaScript. The game must work on GitHub Pages without a server. Put everything in one game_name.html file. This will be stored in my student portfolio repo and linked to my portfolio site.